

Modular extension of the application

Platform, product modules, per-club assignment, and GitHub branch plan. Includes the first step: modular extension capabilities.

Field	Value
Document type	Architecture and delivery plan
Application	LTF License Manager
Baseline	v0.8.0 on feature/ui-refresh-and-designer
First platform release	v0.9.0 — module entitlements
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Status	Design only — no implementation in this document
Audience	Superuser, product owner, implementer

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16. Summary

1. Purpose

This document describes how LTF License Manager can grow as a platform with named modules, without turning Core into a plugin marketplace. Superuser operations (the ops console shipped in v0.8.0) remain the unlock desk. Federation work (LTF Admin, Club Admin) stays separate from product entitlements.

The first delivery is not a calendar, shop, or tournament. It is the modular capability itself: product codes, install entitlements, per-club assignment, API enforcement, and hidden navigation. Every later module is developed on its own GitHub branch.

2. Baseline (v0.8.0)

Core already includes:

- Members, clubs, LTF licenses and cards, print.
- LTF Finance: orders, invoices, payments, club fees billed from the federation to clubs.
- Member transfers, club-admin assignment by LTF Admin, Club Admin rubber-band assignment of home-club members.
- Ops console for Django superusers: sessions, health, security, users, queries, translations, jobs, audit.

New work must never activate an LTF license unless the line is a real license. Club-fee invoices already follow that rule; module invoices must as well.

3. Naming

Do not call a software entitlement a “license.” That word already means a member’s yearly LTF license.

Use	Do not use
Module	Software license
Entitlement / product code	License key (ambiguous)
Unlock / assign to club	License the club

In the ops UI: Modules and product code. Never “software licenses.”

4. Two switches

4.1 Install entitlement

The superuser pastes a signed product code on ops. The payload lists modules, expiry, optional caps, and an install identifier. If this running copy did not buy Kyorugi, nobody — no club, no LTF user — can open it.

4.2 Club assignment

After the install is entitled, Superuser (later optionally LTF Admin) turns the module on or off for named clubs. Example: Club management is entitled for the federation, on for Club A, off for Club B. Club B keeps today's LTF license flow only.

APIs return 403 when a module is locked. Navigation and routes hide locked modules. Federation roles still decide who may act inside an unlocked module. Unlocking Club management does not grant LTF Admin powers.

4.3 Who turns what on

- Superuser: paste product code; see module status; assign modules to clubs.
- LTF Admin (if delegated): assign already-entitled modules to clubs. Cannot invent a module the install did not buy.
- Club Admin: sees Event calendar, Club management, shop, or “host this tournament” only if that club is assigned. Never enters product codes.

5. What stays Core

Not a paid module:

- Auth, members, clubs, LTF licenses and cards, print.
- LTF Finance books and federation club fees (federation invoices clubs).
- Transfers and existing club-admin assignment.
- Ops console (always available to is_superuser).

Modules add capabilities beside Core. They do not replace it.

6. Shared engines

Some capabilities are libraries other modules sit on. They are not sold alone. They ship on the same branch as the first module that needs them, or on a thin core branch merged first.

Engine	Used by
Event	Federation calendar, club calendar, Kyorugi, Poomsae
Inventory / shop	Federation inventory, Club inventory
Tournament core	Kyorugi and Poomsae packs: registration, categories, fees, accreditation, results, external participants

Kyorugi and Poomsae are separate entitlements on one tournament spine. Federation shop and club shop are two faces of one inventory engine. Do not copy-paste two full apps.

7. Module catalog

7.1 Install-wide

A product code unlocks the copy. There is no per-club row, except host flags later.

Module	Purpose	Per club	Notes
Federation inventory	LTF sells goods to clubs; federation stock	No	Install-wide shop
Event calendar (federation)	Federation-published dates	No	On whenever calendar is entitled
Tournament core	Shared tournament spine	No	Implied if either sport pack is entitled
Kyorugi tournament	WT kyorugi: weights, weigh-in, brackets, scoring, protests, external registration	Host clubs only	Visitors are external, not Club Admins
Poomsae tournament	WT poomsae: recognized / freestyle, pair / team, draws, cutoff, scoring	Host clubs only	Same hosting model as Kyorugi

7.2 Per club

Entitled at install, then on or off per club.

Module	Purpose	Who uses it
Club management	Club → member: membership, dues invoices (not LTF licenses)	Club Admin of assigned clubs
Club calendar	That club's events on the shared Event engine	Club Admin of assigned clubs
Club inventory	Club sells goods to members; club-held stock	Club Admin; members buy
Host Kyorugi / Host Poomsae	Only clubs that organize a tournament	Host club + LTF as organizer

External clubs and fighters register as external participants. They do not need Club management or Club inventory. Only the host (LTF or a host club) needs the tournament module.

8. Money flows

Keep these invoice stories distinct. Same Order / Invoice / Payment plumbing is acceptable if every line has an explicit type.

Flow	Seller → buyer	Activates LTF license?
LTF licenses	Federation → Club (member)	Yes, after payment reception.
Club fees (Core today)	Federation → club	No
Club management membership	Club → member	No
Federation inventory	Federation → club (goods)	No
Club inventory	Club → member (goods)	No
Tournament entry	Organizer → registrant	No (fee, not a SKU unless merch is sold later)

Merchant of record: federation shop uses LTF banking / Stripe / Payconiq; club shop uses that club's account where IBAN already exists. Decide this before implementing shops.

SKU and license type must stay different objects. Club management is membership; club inventory is goods. Toggles stay independent.

9. How the domains relate

9.1 Event calendar and tournaments

Event calendar is the light layer: dates, venue, visibility, optional “this event is a tournament.” Tournament modules are heavy events on that calendar: categories, registration, payments, brackets or poomsae draws, operations, results.

One Event record with type calendar / kyorugi / poomsae. Do not build two calendars and two registration desks.

9.2 Tournaments: one core, two sports

Shared: event, venue, timetable, organizer (LTF or host club), registration windows, fees, categories, accreditation, officials, results, audit.

External registration: public or token flow for clubs and fighters not in this install — lightweight external club and athlete, documents, category entry, payment. Do not force Club Admin. After the event they can stay external or be invited into Core as members.

Kyorugi (WT): weight classes, weigh-in / random weigh-in, seeding, single-elim / repechage / best-of, electronic scoring hooks, penalties, video replay, protests, medical.

Poomsae (WT): recognized vs freestyle, individual / pair / team, compulsory poomsae draw, cutoff / semi / final, accuracy vs presentation, non-weight divisions.

World Taekwondo rules change. Put rule packs (categories, scoring, round structure) in versioned data, not only in hard-coded branches.

9.3 Inventory

One stock-and-sales engine, two sellers:

- Catalog: item, SKU, variant (size/color), unit, tax.
- Stock: on-hand, reserved, location (federation warehouse vs club storeroom).
- Documents: simple sale plus stock movement first; delivery notes later if needed.

Federation inventory: LTF Finance catalog; clubs place orders; stock in federation locations. Club inventory: Club Admin catalog for that club only; members buy; no cross-club stock unless consignment is added later. Do not start with consignment.

Inventory can attach to events later (kits, merch). First version is a standalone shop. A tournament fee stays a registration fee, not a SKU, unless merch is sold next to entry on purpose.

10. Ops console — Modules

New area on `/{{locale}}/dashboard/ops`:

- List of modules: on/off, expiry, last validated.
- Enter product code (signed payload).
- Club matrix for every per-club module.
- Every change in the ops audit log.

Show status (active / expired / invalid) without displaying the secret. Codes live in the database, are rotatable, and are not a magic string in .env.

11. First step — modular extension capabilities

This is the whole “can we extend the app?” layer. Nothing user-facing except ops. Until this is merged, do not start calendar, shops, or tournaments on a branch that assumes flags exist.

11.1 GitHub branch

feature/module-entitlements

Branch from the current release line (v0.8.0 / feature/ui-refresh-and-designer, or main only if 0.8.0 is merged first). Do not pile this onto unrelated UI work.

11.2 Version

v0.9.0 — platform release, not a sport or shop release.

11.3 What ships

- Registry of stable module ids, for example: club_management, event_calendar, inventory_federation, inventory_club, tournament_kyorugi, tournament_poomsae.
- Install entitlements from a signed product code (modules, expiry, optional caps, install fingerprint).
- Per-club assignment rows.
- IsModuleEntitled and club-scoped checks on APIs.
- Frontend: entitlements from /api/auth/me/ or a small /api/modules/.
- Ops: Modules page, club matrix, audit.
- Prove hide/show with a dummy or “coming soon” module. No calendar, shop, or tournament yet.

Module id	Scope	Sold as
club_management	Per club	Club management
event_calendar	Install + per-club calendar	Event calendar
inventory_federation	Install-wide	Federation inventory
inventory_club	Per club	Club inventory
tournament_kyorugi	Install; host clubs	Kyorugi tournament
tournament_poomsae	Install; host clubs	Poomsae tournament

11.4 Enforcement

API 403 plus hidden nav. A club with Club management off must not hit club-membership invoice endpoints even with a crafted URL. Public tournament registration URLs must check that the event exists and the host module is on.

11.5 Product codes

Signed tokens verified offline are enough for a private federation app. Optional later: check for revocation. Bind to hostname or a generated instance id so a code cannot be copied to a second install without intent.

12. GitHub branch plan

One GitHub branch per step, including the first step. Own PR, changelog, tests. Do not merge to main unless asked.

#	Branch	Ships	Depends on
0	feature/module-entitlements	Modular capabilities: codes, entitlements, club matrix, 403, hidden nav. Version 0.9.0.	v0.8.0 ops console
1	feature/module-event-calendar	Event engine + federation calendar + per-club club calendar	Branch 0
2	feature/module-club-management	Club membership / dues (club → member invoices)	Branch 0; invoice types from Core
3	feature/module-inventory-federation	Inventory engine + federation shop (LTF → clubs)	Branch 0
4	feature/module-inventory-club	Club shop face + per-club stock	Branch 0 and 3 (same engine)
5	feature/module-tournament-core	Tournament core + external club/fighter registration	Branch 0 and 1 (events)
6	feature/module-tournament-kyorugi	WT kyorugi pack	Branch 5
7	feature/module-tournament-poomsae	WT poomsae pack	Branch 5

After branch 0, later branches register their module id. The module stays locked until a product code says otherwise.

Fewer PRs if needed: branches 3 and 4 can be one feature/module-inventory with federation first and club shop behind the per-club flag. Branches 6 and 7 stay separate entitlements even if developed in series.

13. Why this delivery order

- 0 — Entitlements: you cannot sell or hide anything without it.
- 1 — Calendar: smallest new domain; tournaments hang off Event.
- 2 — Club management: money; reuse invoices carefully.
- 3 — Federation shop: you already invoice clubs.
- 4 — Club shop: member checkout and per-club stock.
- 5 — Tournament core and external registration.
- 6 then 7 — Kyorugi, then Poomsae (rule packs as data).

You can sell calendar plus club management without waiting for full WT tournament software, and still grow into Kyorugi and Poomsae without a second architecture.

14. Rules on every branch

- Superuser is not LTF Admin is not module entitlement.
- Club-on modules: Club A can have Club management; Club B cannot.
- Visitors to a tournament are external; they are not granted Club Admin.
- Goods and dues never activate LTF licenses.
- SKU is not a license type.
- No plugin folder drop-ins; LTF owns every module.
- GDPR for external fighters (minors, photos) is a tournament-core concern, not shop v1.
- Live scoring hardware (for example DAEDO) is a later integration, not v1 of Kyorugi.
- VAT on goods versus sporting services, if needed in Luxembourg: a tax code on the item, not a new app.

15. Risks to decide later, not in step 0

- Payments: who is the merchant — club or federation — for club shop and club dues.
- Negative stock: deny versus allow-backorder, per seller.
- Club-level tournament hosting versus LTF-only organizers.
- Whether LTF Admin may assign modules without Superuser.
- Consignment of federation stock to clubs.

16. Summary

Extend the project as a platform: v0.9.0 on feature/module-entitlements delivers modular capabilities (product codes, install entitlements, per-club matrix, API and UI enforcement). Then one GitHub branch per module: event calendar, club management, federation inventory, club inventory, tournament core, Kyorugi, Poomsae.

Calendar, club management, two inventory faces, and two WT tournament packs sit on a shared Event, Inventory, and Tournament spine. Superuser ops remains the unlock desk. This document is design only; implementation starts when that work is requested, beginning with feature/module-entitlements.